

BUSINESS CASE

GENERIC DIGITAL SIGNING ONLINE TOOL (GDSOT)

1. EXECUTIVE SUMMARY

This Business Case proposes the implementation of a Web-Based Data Import, Integrity Validation, and Digital Signature System under the supervision of the National Informatics Centre (NIC).

The system will securely import structured data from heterogeneous relational databases through secure connectivity interfaces, process and store it within a controlled local environment, apply backend digital signatures to ensure integrity and authenticity, and export digitally validated datasets in Excel and JSON formats.

The project introduces cryptographic validation into government data workflows, strengthening data governance and operational trust while serving as an internship-level proof of concept.

2. BUSINESS PROBLEM

Government departments routinely exchange structured data across databases and operational systems. While relational databases provide reliable storage across departments (PostgreSQL, MySQL, Oracle, etc.), there is no standardized backend mechanism, there is currently no standardized backend mechanism to ensure that datasets remain tamper-evident and verifiable once extracted from their source environment.

This creates risks including:

- Unauthorized data modification
- Integrity compromise during data transfer
- Lack of verifiable authenticity
- Reduced confidence in shared datasets

A structured integrity validation framework is therefore required.

3. Strategic Alignment

The proposed solution aligns with NIC's mandate to promote secure digital governance and strengthen institutional data integrity controls. By integrating backend digital signature validation into structured data workflows, the project enhances accountability, traceability, and trust in government data exchange processes.

This initiative supports the broader objective of establishing secure and standardized data handling frameworks within government systems.

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4. ANALYSIS

Develop a centralized web-based system that imports structured data from multiple relational database systems, applies backend digital signatures, and enables verifiable export of datasets.

Advantages:

- Strong cryptographic integrity assurance
- Tamper-evident validation
- Verifiable authenticity of exported datasets
- Standardized backend validation framework
- Scalable for future expansion
- Aligned with secure digital governance objectives
- Feasible within internship scope

Disadvantages:

- Requires technical implementation effort
- Requires secure digital signature key management

Assessment:

This option provides measurable integrity assurance, aligns with NIC governance objectives, and balances security, feasibility, and scalability. It effectively addresses the identified risks without introducing disproportionate cost or complexity.

5. PROPOSED SOLUTION

The proposed solution is the implementation of a Web-Based Data Import, Integrity Validation, and Digital Signature System designed strengthen integrity within heterogeneous database-driven government workflows.

The system will establish secure connectivity with supported relational databases to import structured datasets into a controlled local environment. Once imported, the data will be processed and stored securely, after which a backend digital signature mechanism will be applied using cryptographic validation techniques.

The digital signature will ensure that:

- The integrity of the dataset is preserved
- Any unauthorized modification can be detected
- The authenticity of the exported data can be verified

The implemented digital signature mechanism ensures cryptographic integrity and system-level authenticity. Future enhancements may incorporate per-user key infrastructure to provide stronger non-repudiation capabilities.

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The system will provide export functionality in widely used formats such as Excel and JSON, enabling operational usability while maintaining integrity assurance.

This approach introduces a standardized, scalable, and cryptographically secure framework for data validation within government data processing environments.

6. Expected Benefits

Implementation of the proposed Web-Based Data Import, Integrity Validation, and Digital Signature System will deliver the following benefits:

Enhanced Data Integrity Assurance

Backend digital signature mechanisms will ensure that imported and exported datasets remain tamper-evident and cryptographically verifiable.

The system provides a reliable method to confirm that data was validated and digitally signed by the authorized system environment.

Reduced Operational Risk

By enabling detection of unauthorized modifications, the system minimizes risks associated with data manipulation during transfer or processing.

Standardized Integrity Validation Framework

The solution introduces a consistent and repeatable backend validation mechanism across heterogeneous database-driven workflows.

Strengthened Trust in Data Exchange

Inter-departmental data sharing becomes more reliable through verifiable integrity controls.

Alignment with Secure Digital Governance Objectives

The initiative supports NIC's mandate to enhance secure, accountable, and transparent data management practices.

7. COST CONSIDERATION

The proposed system is developed as part of an internship initiative and does not require dedicated budget allocation. The implementation utilizes

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existing infrastructure, development tools, and database environments available within the organizational setup.

No procurement of commercial software or additional hardware resources is required. The project therefore represents a low-cost, low-risk initiative focused on strengthening data integrity assurance without financial impact on operational budgets.

8. RISK AND MITIGATION

The implementation of the Web-Based Data Import, Integrity Validation, and Digital Signature System involves certain technical and operational risks. Appropriate mitigation strategies are defined to manage these risks effectively.

1. Cryptographic Implementation Complexity

Risk: Backend digital signature mechanisms require careful handling of hashing algorithms, key management, and verification processes.

Mitigation: Utilize standardized and well-documented cryptographic libraries, conduct controlled testing, and perform code reviews under technical supervision.

2. Database Integration Challenges

Risk: Variations in database schemas and data structures may introduce mapping and transformation challenges during import. Differences in vendor-specific data types and connector protocols may require structured normalization logic.

Mitigation: Implement structured data validation rules, conduct schema compatibility checks, and perform phased testing with representative datasets.

3. Digital Signature Key Management Risk

Risk: Improper storage or handling of private keys may compromise system security.

Mitigation: Store keys in a secure and controlled environment, restrict access privileges, and implement secure key handling practices.

4. Limited Access to Production-Level Data

Risk: Testing may be limited to sample datasets, reducing exposure to real-world complexity.

Mitigation: Use representative structured datasets and perform validation under supervised testing scenarios.

5. Internship Timeline Constraints

Risk: The fixed project duration may limit advanced feature enhancements.

Mitigation: Prioritize core integrity validation features and adopt a phased implementation approach.

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9.IMPLEMENTION AND APPROACH

The project will be executed in a structured and phased manner over a six-month internship period.

Phase 1 – Requirement Analysis and System Design

Define functional requirements, system architecture, data flow, and digital signature implementation framework.

Phase 2 – Data Import and Processing Module Development

Develop secure multi-database connectivity modules using database drivers and abstraction layers to support heterogeneous relational database systems.

Phase 3 – Backend Digital Signature Implementation

Integrate cryptographic logic to generate and apply digital signatures for dataset integrity validation.

Phase 4 – Export Module Integration

Enable export of integrity-protected datasets in Excel and JSON formats.

Phase 5 – Testing and Validation

Conduct functional testing, integrity verification checks, and controlled scenario validation.

Phase 6 – Documentation and Final Demonstration

Prepare technical documentation and present system functionality within the NIC use-case context.

10. RECOMMENDATION

Based on the identified integrity challenges, strategic alignment with secure digital governance objectives, and evaluation of available options, it is recommended to proceed with the implementation of the proposed Web-Based Data Import, Integrity Validation, and Digital Signature System.

The solution provides cryptographic integrity assurance, strengthens trust in structured data exchange, and establishes a scalable validation framework within PostgreSQL-based government workflows. Given its low-cost, internship-based execution model, the project presents minimal financial risk while delivering measurable governance value.